

HealthConnect Experience Lab — Week 7 Executive Report

Analytics Testing, KPI Validation & Feature Refinement | Author: Brian Kariuki

1. Executive Summary & Audit Verification

Week 7 focused on rigorous testing, statistical stability validation, and cross-track alignment. The baseline dataset (5,000 records) was programmatically audited via assertion checks to confirm zero drift from Week 5/6 metrics. Validated segment outputs were exported to `HealthConnect_Feature_Engineered_Segments_Validated.csv`.

2. Programmatic KPI Audit Matrix

Metric / Operational Driver	Calculated Value	Audit Standard	Status	Execution Output & Findings
Baseline No-Show Rate	48.46%	48.46% (2,423 / 5,000)	PASSED	Assert condition passed with 0.00% delta.
SMS Reminder Attendance	49.60%	Highest vs No Reminder (42.68%)	PASSED	SMS outperformed Email (46.53%) and WhatsApp (44.60%).
Lead Time Gap (No-Show)	34.53 Days	vs Attended (24.52 Days)	PASSED	10.01-day mean extension for non-attending cohort.

3. Statistical Segment Stability & Lead Time Refinement

- **Lead Time Bracket Segmentation:** Cross-tabulation confirms a non-linear drop-off cliff. No-shows stay at ~37.21% for the 8-21 day bracket but jump significantly for longer booking intervals.
- **Data Science Artifact Handoff:** Exported `HealthConnect_Feature_Engineered_Segments_Validated.csv` containing binned features (lead_time_bracket, age_group, high_wait_flag) to reduce false-negative rates in downstream ML pipelines.
- **Production Readiness:** Execution logs confirm complete pipeline stability and zero missing value propagation across all 5,000 engineered rows.